



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

CSA1017 – SOFTWARE ENGINEERING

INDIVIDUAL REFLECTION REPORT

INTELLIGENT DISASTER EARLY WARNING AND EVACUATION MANAGEMENT SYSTEM

NAME: GANTA RAM CHARAN

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ROLE: GIS Mapping and Evacuation Route Planning

DESIGN / DEVELOPMENT DECISION: I worked on the GIS-oriented risk map, hazard zones, shelter markers, emergency facilities, road conditions, and evacuation-route comparison. I decided that route selection should consider safety, travel time, distance, accessibility, closures, and hazard exposure instead of using distance alone.

CHALLENGE FACED AND SOLUTION: The main challenge was designing a meaningful mapping and evacuation workflow without live geographic, traffic, or road-closure data. I addressed this by defining a simulated geographic environment, adding a map fallback mode, and clearly labelling route recommendations as prototype decision support.

LEARNING GAINED: I gained practical knowledge of spatial data, map layers, route scoring, accessibility considerations, and the relationship between GIS and emergency decision-making. I learned that the shortest route may not be the safest route, particularly when roads are blocked, hazards are spreading, or citizens have mobility and medical constraints.

SDG RELEVANCE: The project relates to SDG 11 because GIS-based risk visibility and evacuation planning can support safer and more resilient settlements.

COURSE OUTCOME CONTRIBUTION: My work supports the course outcome related to GIS, algorithmic problem-solving, and application design by connecting spatial information with evacuation decisions.

POSSIBLE IMPROVEMENT: Future work should integrate authoritative map data, live traffic and closure feeds, accessible routing, and field testing with emergency responders.